Analysis code for CogSci 2026: ‘Inferring Arithmetic Skill from Speed and Accuracy’

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Study 1

H1 (Difficulty judgments): $\beta = 0.45$, t(911.69) = 22.18, p < .001.

H2 (Competence inferences from a single outcome): $\beta = 0.86$, t(418) = 34.72, p < .001, Adjusted $R^2 = 0.74$.

H3–H4 (Model accuracy and comparison):

- Main (Bayesian) Model: $\beta = 0.94$, t(418) = 58.29, p < .001, Adjusted $R^2 = 0.89$.
- Threshold: Adjusted $R^2 = 0.39$.
- Anchor: Adjusted $R^2 = 0.59$.
- BIC: $BIC_{\text{Bayes}} = 16{,}063.72$; $BIC_{\text{Threshold}} = 19{,}411.92$; $BIC_{\text{Anchor}} = 18{,}259.96$.

Robustness (bottom 30% of arithmetic performers):

- Perceived vs. objective difficulty: $\beta = 0.93$, t(303) = 44.51, p < .001.
- Conditional predictions: $\beta = 0.85$, t(448) = 34.03, p < .001, Adjusted $R^2 = 0.72$.

Study 2

H1 (Difficulty judgments): $\beta = 0.39$, t(313.06) = 16.85, p < .001.

H2a (Fast accurate answers are competence cues): b = 0.51, z = 2.15, p = 0.031.

H2b (Fast inaccurate answers are competence cues): b = 0.71, z = 2.63, p = 0.009.

H3 (Competence inferences from a single outcome): $\beta = 0.89$, t(98) = 19.47, p < .001, Adjusted $R^2 = 0.79$.

H4a (Participants treat faster accurate answers as competence cues): b = 0.17, z = 1.53, p = 0.125.

H4b (Participants treat faster inaccurate answers as competence cues): b = 0.15, z = 1.62, p = 0.104.

Robustness (bottom 30% of arithmetic performers):

- Perceived vs. objective difficulty: $\beta = 0.32$, t(88.04) = 8.16, p < .001.
- Conditional predictions: $\beta = 0.88$, t(98) = 18.73, p < .001, Adjusted $R^2 = 0.78$.
- Trial-level analysis: b = 1.95, z = 46.19, p < .001.